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Solution

Machine Learning based Real-time Fall Detection
and Emergency Response System for the elderly.

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Abstract—In this study, we investigate the development of a fall detection and emergency response system for the elderly using machine learning. Our study analyses motion sensor data to evaluate the effectiveness of machine-learning methods for fall detection. We also take into account the prospect of using an alarm mechanism. Insights from our research can be used to help identify falls in older people as well as respond to emergencies.

Keywords— Fall detection, sensors, machine learning

I. INVESTIGATION-INTRODUCTION

Elderly people experience the majority of injuries, disabilities, and fatalities from falls, which puts a significant financial burden on individuals, families, and social public health systems (Leggett, C. et al, 2017). Each year, a multitude of individuals, primarily the elderly encounter falls resulting in major injuries that can lead to fatalities or extremely serious impairments. The senior population is more prone to falls, with 28–35% of those 65 and older fall each year and that number rising to 3–22% for those 70 and older. Furthermore, due to variables including reduced bone mineral density, elderly ladies are more likely to sustain hip-related fractures after falls. (World Health Organization, 2008). Additionally, due to their decreased stride and weaker reflexes, people with disabilities are also at a relatively high risk of falling due to disorders like post-operative conditions and Parkinson's disease (Yoshida-Intern, S., 2007). Consequently, if we are to solve this global health concern, an effective fall detection and alarm system is necessary. An abrupt instability in pace brought on by an accidental trip, external impact, or several locomotor irregularities may cause a subject to lose balance and fall to the ground. Falling is a problem of postural instability in human gait studies. Proprioceptive awareness, dynamic stability, and neuromuscular engagement are all components of the behavioural learning technique known as "fall recovery." Humans can only recover from falls to a certain extent, and this capacity varies with age, gender, terrain, tiredness, injury, and posture. Due to ageing-related neuromuscular deterioration, this ability declines (Semwal, V. B., 2022). This study aims to investigate how machine learning techniques can be applied to older individuals' fall detection and emergency response in real time. A prompt response can considerably lower the risk of injury and mortality. This study attempts to determine how machine learning can be used and prepared to address this issue given the high potential it offers in this field.

II. LITERATURE REVIEW

In fall detection or prevention systems that use machine learning-based techniques, common algorithms include Naive Bayes, KNN, SVM (Support Vector Machine), random forest, fuzzy logic, Hidden Markov Mode (HMM), etc. For instance, Howcroft et al, (2017) looked into the impact of combining two wearable sensors to predict fall dangers. The two wearable sensors used were an accelerometer and pressure-sensing insoles. Furthermore, the head, right shank, foot, and left pelvis were all equipped with their own individual accelerometers. SVM (Support vector machine), neural networks, and naïve Bayesian were the three models utilized. The findings demonstrated that gait evaluation using various sensors, such as hybrid readings of the left shank, head, and posterior pelvis accelerometers, may provide the greatest information for predicting falls. However, while many studies have focused on the technical aspects of fall detection, there is a lack of research on the practical applications of these systems. The remedy for the creation and evaluation of wrist-worn fall detection has been provided by (de Quadros, T. et al, 2018). To develop the most effective fall detection method, a variety of signals along with directional components were merged with machine learning algorithms. The sensors that were used were a magnetometer, gyroscope, and accelerometer. The signals employed were acceleration, displacement, and velocity, and the directions were perpendicular as well as horizontal. For the project, information on 22 volunteers' fall and non-fall movements were gathered. The accuracy, specificity, and sensitivity were all 99.0% using machine learning approaches. When further examined using threshold approaches, the study's findings additionally showed 91.1% accuracy, 86.5% specificity, and 95.8% sensitivity. Comparatively, fall detection systems based on machine learning outperform threshold-based systems by training classifiers with extracted characteristics. Similarly, it was suggested that research be done to identify falls in older people who are indoors. This was a cloud computing, big data, and low-power network of wireless sensors based on an IOT (Internet of Things) system. It was implemented using a wearable 6LoWPAN gadget equipped with a 3D-axis accelerometer that could gather movement data from elderly persons. A decision tree-based model was used to analyze the sensor's reading (Yacchirema, D. et al, 2018). The system promptly reacts to the detection of a fall by activating an alert and sending alerts. Finally, the services will be delivered using cloud technology. Every

time a fall is detected, the system offers a service that makes use of this data to develop machine learning models. The work demonstrated a high degree of effectiveness in producing outcomes that were both precise and accurate. Likewise, (Gharghan, S. K. et al, 2018) describes a method that pinpoints the location of an older person who has fallen while travelling inside a building or other enclosed space. After using a sensor-based fall detection method to identify when falls occur, an artificial neural network was used to determine the position. This method is interesting; however, it relies on artificial neural networks, which might not be the best approach for real-time fall detection. An HMM-based method for fall detection utilizing a single 3-axial accelerometer was created by Yu et al. (2018). To train the HMM, acceleration signals were evaluated on the basis of Gaussian distributions for hidden states. A multi-sensor study was conducted by Boutellaa. et al. (2019) in which he proposed a classification model based on a KNN covariance matrix from which the signals of the activity could be determined. For this study, the readings of an accelerometer, gyroscope and magnetometer were used. In more recent studies, (Kulurkar, P. et al, 2023) offered a framework for edge computing that makes use of real-time monitoring with LSTM networks, which are nothing more than recursive neural networks that are good at handling time-related information and can recall previous recollections of information. The effectiveness of machine learning-based fall detection and prevention systems have been highlighted in the reviewed studies, especially when numerous sensors and machine learning algorithms are integrated. Nonetheless, the need for practical ways to help older people who have fallen remains. To enable fast response and assist the elderly in the case of a fall, one potential option is to incorporate alarm systems within the sensors utilized in these systems. Such solutions, which can ultimately increase the safety and quality of life for elderly individuals, need to be identified and developed.

III. FINDINGS

Studies have demonstrated that fall detection and prevention systems using machine learning-based techniques are beneficial. Typical algorithms used in these systems include KNN, SVM, Naive Bayes, HMM, random forest, and fuzzy logic. The best data for forecasting falls can be acquired when many sensors are employed, like a combination of accelerometers and pressure-sensing insoles. On the practical applications of these systems, there is, however, a dearth of research. Wearable sensors were used in a study that revealed machine learning algorithms had a 99.0% accuracy rate for detecting falls. Other research has focused on identifying falls in elderly people who are indoors, with an IoT-based system employing big data, cloud computing, and low-power wireless sensor networks proving effective in delivering precise and accurate results.

Another approach entails utilizing an artificial neural network to locate an elderly person who has fallen. Another solution relies on a single 3-axial accelerometer and a fall detection system based on HMM. In a recent edge computing architecture, fall detection and prevention are aided by real-time surveillance with LSTM networks.

Alarm systems can be built into the sensors used in these systems to improve the safety and quality of life for elderly people. Overall, machine learning-based fall detection and prevention systems are promising, but there is still a need to identify and create workable ways for supporting the elderly in the event of a fall.

IV. CONCLUSION-1500 WORDS

It is evident from the literature review and findings that machine-learning approaches hold significant promise for developing efficient fall detection and prevention systems for senior citizens. Wearable sensors and a variety of machine-learning algorithms have been combined to successfully detect falls with high sensitivity and specificity. Systems based on the Internet of Things and cloud computing have also been put in place to offer real-time monitoring and emergency response. There is still a need for usable apps and solutions to help older people who have fallen, even if the technological parts of fall detection have received a lot of attention. To enable quick response and help the elderly in case of a fall, alarm systems included into the sensors may be an option. Overall, machine learning-based fall detection systems present promising chances to improve elderly people's safety and quality of life. The healthcare sector and society at large may be significantly impacted by additional research and development in this field.

V. SOLUTION- METHODOLOGY

The goal of this study was to provide a software solution that would categorize simulated falls and daily living activities. It is extremely important to be able to effectively discern between falls and normal daily activities in order to give individuals who might need medical assistance immediate attention. Different machine learning techniques, such as SVM, Random Forest, Neural Networks, KNN, Gradient Boosting, and Decision Tree, were used to solve this issue. A total of 3060 occurrences from 17 participants' average of 5 repetitions of 36 different movements were included in the dataset that was used to train and test the models. Each movement was captured by five sensors, including a 3-axis accelerometer, gyroscope, and magnetometer. The dataset comprises 20 different types of falls and 16 activities of daily living. Several metrics, including accuracy, F1 score, precision, recall, and AUC-ROC, were employed to assess the models' performance. Since they offer a thorough evaluation of how effectively the models can distinguish between falls and activities of daily living, accuracy and AUC-ROC were the key areas of focus.

VI. ACTIVITIES UNDERTAKEN

The data was gathered using a Python script that first took the directory path to the folder containing all of the test data. The script then went through all the activity folders and test folders to extract data from the sensor files for each test after obtaining the volunteer's gender, age, height, and weight from a text file inside each folder.

For each sensor file it read in with pandas, the script added columns for the volunteer's activity, test number, and sensor name as well as columns for the volunteer's gender, age, height, and weight before appending the information to a collection of data frames. After gathering the data, we filtered those rows in the sensor column that had the phrase "Unknown sensor," making the appropriate changes to the columns. Activity Type and Activity Number were then separated into two columns from the activity field. The data frame displayed in Fig.1 and Fig.2 was generated.

	Counter	Temperature	VelInc_X	VelInc_Y	VelInc_Z	OriInc_w	OriInc_x	OriInc_y	OriInc_z	Acc_X	Acc_Y	Acc_Z	Gyr_X	Gyr_Y	Gyr_Z	Mag_X	Mag_Y	Mag_Z	Pressure	Roll	Pitch	Yaw	Gender	Age	Height
0	0	0.001160	-0.373230	-0.121765	0.999999	0.000031	0.000516	0.000637	0.028992	-9.330750	...	NaN	Male	27	176										
1	0	0.001115	-0.373789	-0.122840	0.999999	0.000232	0.000402	0.000549	0.027871	-9.344721	...	NaN	Male	27	176										
2	0	0.000427	-0.373657	-0.121826	0.999999	0.000496	0.000478	0.000340	0.010681	-9.341431	...	NaN	Male	27	176										
3	0	-0.001422	-0.374447	-0.119565	0.999999	0.000605	0.000301	0.000519	-0.035548	-9.361172	...	NaN	Male	27	176										
4	0	-0.000427	-0.373535	-0.118042	0.999999	0.000526	0.000348	0.000519	-0.010681	-9.338379	...	NaN	Male	27	176										

Fig.1

Weight	Volunteer	Test	Sensor	Activity Type	Activity Number
67	104	1	Right Ankle Sensor	Daily activities	801
67	104	1	Right Ankle Sensor	Daily activities	801
67	104	1	Right Ankle Sensor	Daily activities	801
67	104	1	Right Ankle Sensor	Daily activities	801
67	104	1	Right Ankle Sensor	Daily activities	801

Fig.2

This new data frame underwent more preprocessing. Using the drop() function, the columns that were unnecessary, such as the "Counter," and the columns with an excessive number of missing values, such as the "RSSI," were eliminated. Following that, the dataset was examined for null values, which were not many. After dropping the rows with null values, a significant amount of data remained, therefore the null rows were subsequently deleted using the dropna() function. The 'object' data types were then changed to 'categorical' to prepare for further encoding. The final data frame had the following layout (Fig. 3).

VelInc_X	float64
VelInc_Y	float64
VelInc_Z	float64
OriInc_w	float64
OriInc_x	float64
OriInc_y	float64
OriInc_z	float64
Acc_X	float64
Acc_Y	float64
Acc_Z	float64
Gyr_X	float64
Gyr_Y	float64
Gyr_Z	float64
Mag_X	float64
Mag_Y	float64
Mag_Z	float64
Pressure	float64
Roll	float64
Pitch	float64
Yaw	float64
Gender	category
Age	int64
Height	float64
Weight	float64
Volunteer	int64
Test	float64
Sensor	category
Activity Type	category
Activity Number	category

Fig.3

Feature scaling was then carried out. The following features were chosen for the feature scaling: Pressure, Roll, Pitch, and Yaw, VelInc_X, VelInc_Y, VelInc_Z, OriInc_w, OriInc_x, OriInc_y, OriInc_z, Gyr_X, Gyr_Y, Gyr_Z, and Mag_X, Mag_Y, Mag_Z. These columns were scaled using the standard scaler. The ordinal encoder was then used to encode the categorical variables Sensor and Activity type. The 'Activity Number' column from the original data frame was added to the resultant data frames, which were then combined into a single data frame. The generated data frame is shown in Fig.4.

#	Column	Dtype
0	VelInc_X	float64
1	VelInc_Y	float64
2	VelInc_Z	float64
3	OriInc_w	float64
4	OriInc_x	float64
5	OriInc_y	float64
6	OriInc_z	float64
7	Acc_X	float64
8	Acc_Y	float64
9	Acc_Z	float64
10	Gyr_X	float64
11	Gyr_Y	float64
12	Gyr_Z	float64
13	Mag_X	float64
14	Mag_Y	float64
15	Mag_Z	float64
16	Pressure	float64
17	Roll	float64
18	Pitch	float64
19	Yaw	float64
20	Sensor	float64
21	Activity Type	float64
22	Activity Number	category
dtypes: category(1), float64(22)		

Fig.4

Additionally, the 'Activity Type' target variable was checked for imbalances. Although there was a tiny imbalance, it had no bearing on the outcome, as shown in Fig. 5.

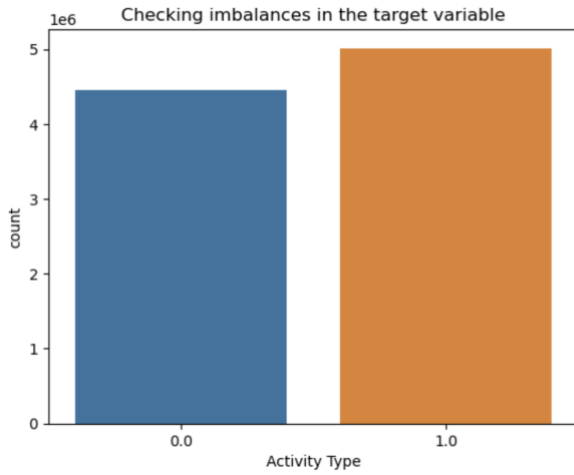


Fig.5

The completed dataset had the following shape (9468875, 23). The dataset was trimmed down to 30000 rows for hyper tuning as there was a significant computational cost. Using 'random_state=42', the rows were chosen at random. Following that, this was divided into 70% train sets and 30% test sets. With 'random_state=92', the final dataset's validation set was likewise chosen at random. The models chosen were Gradient Boosting, SVM, Random Forest, KNN, Neural Networks, and Decision Tree. To determine the ideal parameters, each model underwent hyper-tuning using GridSearchCV. Using these parameters, we trained the model and then evaluated which model produced promising results. AUC-ROC, F1 score, recall, precision, and accuracy were the metrics implemented to evaluate this. We paid close attention to the AUC-ROC scores and accuracy. After the models were tuned and the best parameters were obtained, we then used these new parameters to further train the corresponding models. To do this, the dataset was split again from the original. We took 99999 instances from the dataset with a 'random_state=96'. The resulting dataset was then split again into test and train datasets in order to train the models. The best parameters obtained were,

- 1) SVM
Best hyperparameters: {'C': 10, 'gamma': 'scale', 'kernel': 'rbf'}
- 2) Random Forest
Best hyperparameters: {'max_depth': None, 'min_samples_leaf': 1, 'min_samples_split': 2, 'n_estimators': 150}
- 3) Neural Network
Best hyperparameters: {'activation': 'relu', 'optimizer': 'adam'}
- 4) KNN
Best hyperparameters: {'metric': 'manhattan', 'n_neighbors': 1, 'weights': 'uniform'}
- 5) Decision Tree
Best hyperparameters: {'criterion': 'entropy', 'max_depth': None, 'max_features': None, 'min_samples_leaf': 1, 'min_samples_split': 2}
- 6) Gradient Boosting
Best hyperparameters: {'max_depth': 7, 'n_estimators': 150}

VII. FINDINGS OF THE RESEARCH

After tuning and training six machine learning models, the following findings were observed. Random Forest and KNN worked much better than the other models, with Gradient Boosting not far behind. The accuracy rate for the Random Forest model was 95%, which was the highest accuracy rate obtained. This was followed by KNN and Gradient Boosting, having the accuracy rate of 94% and

93% respectively. The performance of the models can be seen in the plot (Fig.6). The scores for the individual models were as such:

1) SVM

Accuracy: 0.8453
F1 score: 0.8611933602512338
Precision: 0.8204353772509688
Recall: 0.9062126266758985
AUC-ROC: 0.8414716502613533

2) Random Forest

Accuracy: 0.9584666666666667
F1 score: 0.960913482652613
Precision: 0.9577887561753486
Recall: 0.9640586643167369
AUC-ROC: 0.9581152104266317

3) Neural Network

Accuracy: 0.7608
F1 score: 0.7862504468009054
Precision: 0.746282160022618
Recall: 0.8307421161956317
AUC-ROC: 0.8344425021911396

4) KNN

Accuracy: 0.9474666666666667
F1 score: 0.9504527162977867
Precision: 0.9494378493813203
Recall: 0.9514697551457166
AUC-ROC: 0.9472150731372315

5) Decision Tree

Accuracy: 0.9243333333333333
F1 score: 0.9287105081339112
Precision: 0.9267314321529301
Recall: 0.9306980550135331
AUC-ROC: 0.9239333115002477

6) Gradient Boosting

Accuracy: 0.9325666666666667
F1 score: 0.9376098689282961
Precision: 0.9191558834200024
Recall: 0.9568200415434003
AUC-ROC: 0.9310423455786159

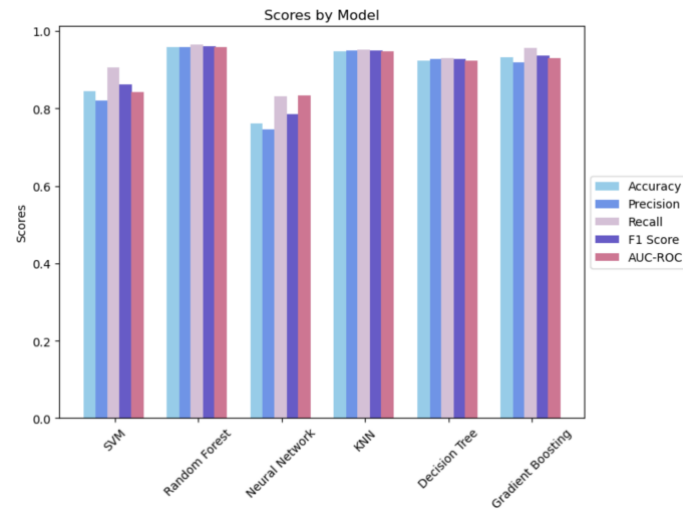


Fig.6

From the accuracy rates we concluded Random Forest, KNN and Gradient boosting classifiers show the highest rates of accuracy. Using the AUC-ROC score we then plotted the ROC (Receiver Operating Characteristic), which shows any binary classifier's diagnostic ability. The AUC (Area Under Curve) shows the measure of performance of any given classification model. The higher the value, the better the performance. Figures 7,8 and 9 represent the ROC curve for the Random Forest (0.99), KNN (0.95) and Gradient boosting (0.98).

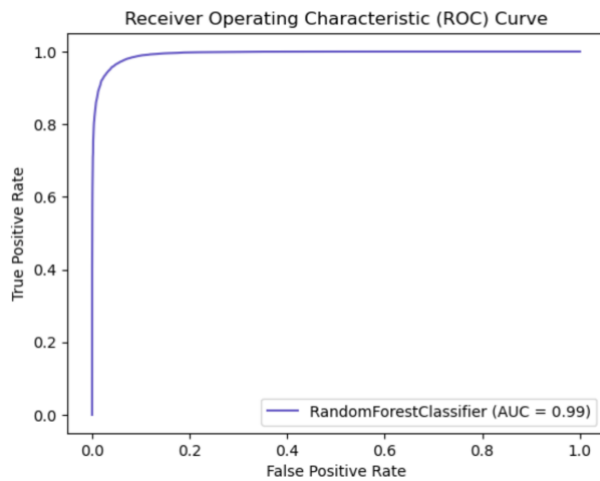


Fig.7 ROC for Random Forest model.

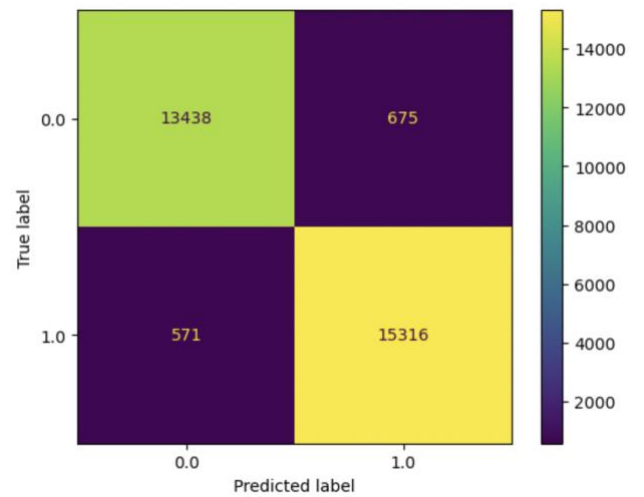


Fig.10 Confusion Matrix for Random Forest.

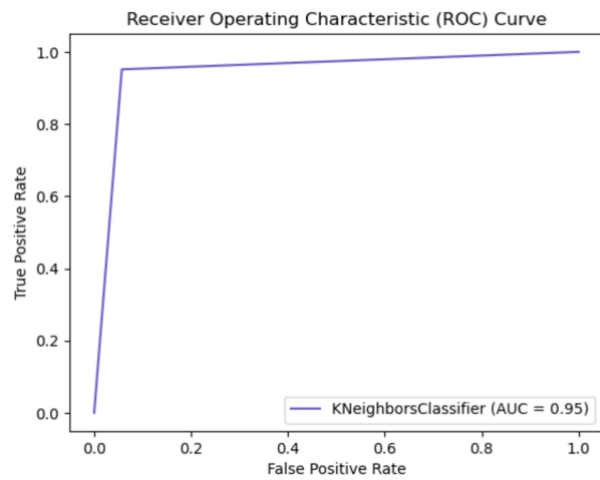


Fig.8 ROC for KNN model.

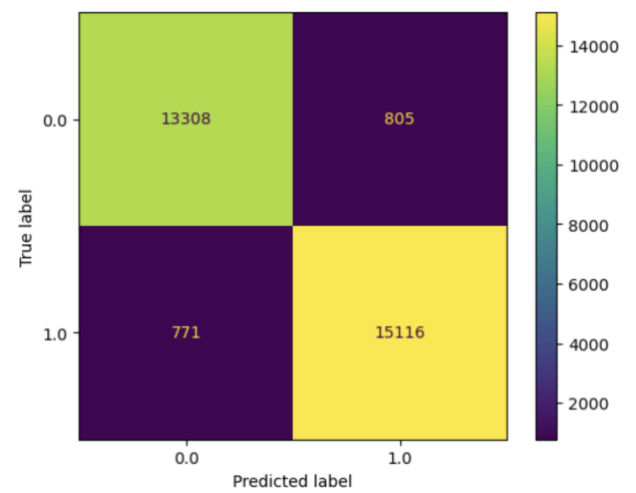


Fig.11 Confusion Matrix for KNN.

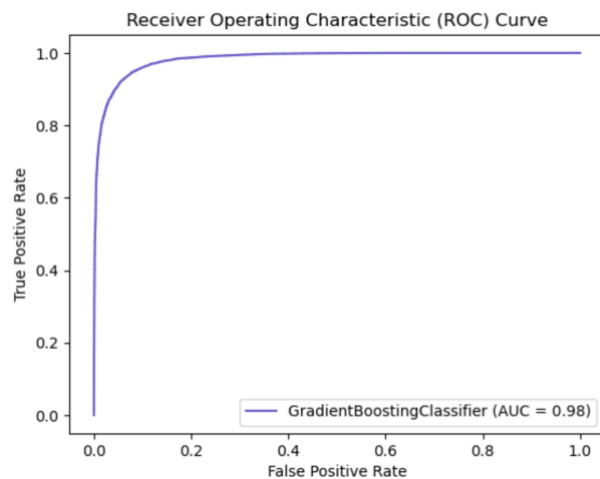


Fig.9 ROC for Gradient Boosting model.

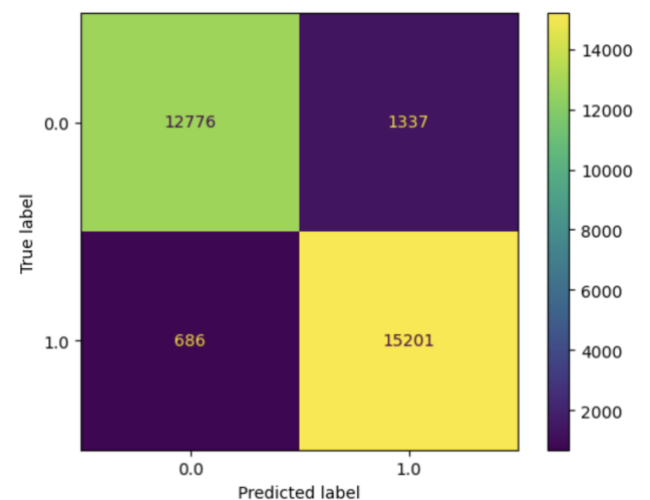


Fig.12 Confusion Matrix for Gradient Boosting.

We also used a confusion matrix to observe how the classifiers predicted true positives, false positives, true negatives, and false negatives as shown in Figures 10, 11, and 12.

Looking at these results we concluded that the Random Forest and KNN classifiers were better. Therefore, using the trained models the validation set of 10 randomly selected instances were put through these two classifiers. The results are shown in Figures 13 and 14.

	Activity Number	Activity
0	802	[0.0]
1	805	[0.0]
2	806	[1.0]
3	808	[0.0]
4	809	[0.0]
5	908	[1.0, 1.0]
6	911	[1.0]
7	914	[1.0]
8	918	[1.0]

Fig.13 Results from Random Forest Classification.

	Activity Number	Activity
0	802	[0.0]
1	805	[0.0]
2	806	[1.0]
3	808	[0.0]
4	809	[0.0]
5	908	[0.0, 1.0]
6	911	[1.0]
7	914	[1.0]
8	918	[1.0]

Fig.14 Results from KNN Classification.

Instances starting with 8** are supposed to be daily living activities and should be represented by a '0.0'. Likewise, instances starting with 9** are fall actions and should be represented by '1.0' which represents a fall has occurred. Here, the Random Forest model results are more accurate with an error of one instance, instance 2. However, the KNN model shows two errors at instance 2 and instance 5.

VIII. FURTHER RESEARCH

The Random Forest model was the most effective classifier in the course of our research that was used to determine whether a fall has occurred or not. It has an accuracy score of 95% and when put to test it predicted the results much more accurately with one exception as compared to the KNN model which had an accuracy rate of 94%. The next step going forward is to develop an alert system to detect the falls for the safety of the elderly.

Using this Random Forest classifier, it will be easier to accurately detect falls and provide alerts to caregivers. The sensors in the alert system can take into account the factors that contribute to the falls, such as gait and balance. For this, elderly people should wear the device at all times. Any drastic change in motion should be picked up by the sensor and trigger the Random Forest model to make a prediction about the occurrence of a fall. If the result is a true positive, the system should then be able to send emergency messages to the caregivers.

Another advantage of the system is that the system can be completely personalized. On collecting the data of the person's daily movements and activities the Random Forest model will adapt to the motions of that specific individual.

IX. CONCLUSION-1511 WORDS

In conclusion, a Random Forest model has the potential to improve the safety and day-to-day life of elderly people.

The accuracy, ROC_AUC, F1 score, recall and precision were the metrics used in determining this. The confusion matrix too showed a higher true positive rate and a low false positive rate. This is important in minimizing false alarms. Detecting and alerting caregivers or emergency services in the event of a fall by integrating the wearable sensor with a smartphone or any other device allows them to respond quicker and reduces the risk of a fatal injury.

With daily use, the personalized system will prove beneficial for the elderly individuals as well as their caregivers. Also, the use of a wearable sensor is a more practical and non-invasive approach to tackle this issue. Overall, the findings of this study show that machine learning demonstrates promising results in this area to detect falls.

REFERENCES

- Boutellaa, E., Kerdjidi, O., & Ghanem, K. (2019). Covariance matrix based fall detection from multiple wearable sensors. *Journal of biomedical informatics*, 94, 103189.
- de Quadros, T., Lazzaretti, A. E., & Schneider, F. K. (2018). A movement decomposition and machine learning-based fall detection system using wrist wearable device. *IEEE Sensors Journal*, 18(12), 5082-5089.
- Gharghan, S. K., Mohammed, S. L., Al-Naji, A., Abu-AlShaeer, M. J., Jawad, H. M., Jawad, A. M., & Chahl, J. (2018). Accurate fall detection and localization for elderly people based on neural and energy-efficient wireless sensor networks. *Energies*, 11(11), 2866.
- Howcroft, J., Kofman, J., & Lemaire, E. D. (2017). Prospective fall-risk prediction models for older adults based on wearable sensors. *IEEE transactions on neural systems and rehabilitation engineering*, 25(10), 1812-1820.
- Kulurkar, P., kumar Dixit, C., Bharathi, V. C., Monikavishnuvarthini, A., Dhakne, A., & Preethi, P. (2023). AI based elderly fall prediction system using wearable sensors: A smart home-care technology with IOT. *Measurement: Sensors*, 25, 100614.
- Leggett, C., Jess, J., & McNamara, R. (2017). A Multi-disciplinary approach to falls prevention in the elderly. *International Journal of Integrated Care*, 17(5), A202.
- Semwal, V. B., & Nandi, G. C. (2015). Toward developing a computational model for bipedal push recovery—a brief. *IEEE Sensors Journal*, 15(4), 2021-2022.
- World Health Organization, World Health Organization. Ageing, & Life Course Unit. (2008). WHO global report on falls prevention in older age. World Health Organization.
- Yacchirema, D., de Puga, J. S., Palau, C., & Esteve, M. (2018). Fall detection system for elderly people using IoT and big data. *Procedia computer science*, 130, 603-610.
- Yoshida-Intern, S. (2007). A global report on falls prevention epidemiology of falls. Geneva: WHO.
- Yu, S., Chen, H., & Brown, R. A. (2017). Hidden Markov model-based fall detection with motion sensor orientation calibration: A case for real-life home monitoring. *IEEE journal of biomedical and health informatics*, 22(6), 1847-1853.